He Min

China | +65-89902569
MHE009@e.ntu.edu.sg | https://qiyuanlanniao.github.io

EDUCATIONAL BACKGROUND

Nanyang Technological University **Singapore**
Master of Science in Signal Processing and Machine Learning 2026.01-2027.07
Chongqing University of Technology **Chongqing, China**
Bachelor of Science in Computer Science and Technology (Honours) 2021.09-2025.06

- **GPA:** 3.96 / 5.0 (**Top 1%**)

PUBLICATION

He, M. (2025). **A Review of Data Fusion and Deep Learning Models for Multimodal Sentiment Analysis.** In *Proceedings of the 1st International Conference on E-commerce and Artificial Intelligence – ECAI* ***(CCF B)***; ISBN 978-989-758-726-9, SciTePress, pages 5-12. DOI: 10.5220/0013189700004568

PROJECT EXPERIENCE

OV-LidarMap: Open-Set Instance-Level 3D Semantic Mapping with Rotating LiDAR and Panoramic Vision 2026.01-Present

First author, submitted to IEEE RA-L **(SCI Q2 TOP)** Supervisor: Prof. Lihua Xie

Project Summary: Addressing the issues of missing spatial reasoning and context explosion in large-scale 3D scenes using LLM (Laser-Driven Model) frameworks, a perceptually driven embodied intelligence agent framework was developed based on Python, PyTorch, and ROS2. By constructing a dynamic scene graph as the agent's world model and employing a multi-resolution spatial attention mechanism, complex long-range navigation in zero-shot environments was achieved with a success rate of 93.18%, while reducing token costs by up to 69.98%.

• Intelligent Agent Architecture: A self-reflective Actor-Critic decision-making brain was constructed using LangChain and LangGraph. A hierarchical chain-based reasoning system was designed to guide the LLM to execute tasks following the logic of "room filtering - local orientation - fine-grained locking". By integrating Qwen2.5-VL and a custom spatial toolkit, the spatial illusion problem common in traditional LLM navigation was solved, improving the long-range navigation success rate by 15.91%.

• Hierarchical World Model: A five-layer dynamic scene graph was constructed. Object instantiation under omnidirectional vision was achieved using SAM2, and adaptive room partitioning was implemented based on a continuous cohomology algorithm, completing the semantic leap from low-level point cloud geometry to high-level abstract space. A geometry-VLM hybrid pruning technique was introduced, combined with KDTree ray detection and online verification of multimodal models, and Open3D was used to process high-frequency point clouds to ensure the physical authenticity of the scene graph's topological connections.

• Multi-resolution Spatial Attention: An innovative 3D stereo eight-neighbor observation model centered on the robot was proposed, transforming the complex coordinate system into a location description that conforms to human intuition. An adaptive resolution prompting mechanism was designed, which provides fine-grained captions only for objects in the focus area and performs semantic summarization on the peripheral and far-field spaces, thereby compressing the accumulated token overhead by nearly 70% while ensuring decision accuracy.

HSM-Net: A Multimodal Diagnostic Framework for Abdominal Trauma Based on Frequency Domain Denoising and LLM Prior 2025.10-Present

First author, intending to submit to Artificial Intelligence in Medicine **(SCI Q2 TOP)** Supervisor: Prof. Xiaofei Zhu

Project Summary: Addressing the challenges of high-frequency speckle noise interference and the cross-modal semantic gap between physiology and imaging in abdominal trauma ultrasound diagnosis, a deep learning framework integrating frequency domain awareness, large model priors, and dynamic decision-making was developed using Python and PyTorch. On real-world clinical datasets, it achieved a binary classification accuracy of 91.87% and a four-level injury grading accuracy of 90.62%, comprehensively outperforming 12 state-of-the-art models, including CardiacCLIP.

• Spectrum-Aware Denoising: To address the challenge of effectively removing ultrasound speckle noise from spatial domain convolution, a spectrum-aware denoising module based on Fast Fourier Transform (FFT) was designed. This module adaptively suppresses noise components using a learnable complex weight matrix, significantly enhancing the model's ability to perceive subtle pathological textures even at low signal-to-noise ratios. A parallel dual-stream encoder, combining Swin-Transformer and DenseNet, was constructed to achieve efficient fusion of global anatomical structures and local damage details.

• LLM Clinical Semantic Injection: Utilizing BERT as a knowledge engine, discrete physiological indicators (blood pressure, heart rate, etc.) were transformed into high-dimensional structured embeddings through Prompt Engineering. A bidirectional cross-modal attention mechanism was introduced to simulate the decision-making process of human doctors verifying imaging findings based on vital signs. Ablation experiments demonstrated that this module improved diagnostic accuracy by 4.32%.
• Dynamic MoE Expert System: A dynamic routing network based on uncertainty gating was constructed, adaptively activating specific expert modules according to the complexity of the damage in the case. For typical samples (0 points, 3 points), a general path is used to reduce overhead, while for difficult samples with ambiguous boundaries (1 point, 2 points), a specific path is activated to enhance the interpretation depth. Experiments show that the system significantly improves the classification accuracy of difficult cases and increases the recall rate of the severe category to 91.87%.

AWARDS & HONORS

• National Competitions: Second Prize in the National Undergraduate Data Statistics and Analysis Competition (2023); Finalist Award in the National Algorithm Elite Competition (2023); Third Prize in the National Team Programming Ladder Competition (2023).
• Comprehensive Honors: National Scholarship (2024); Outstanding Graduate of Chongqing Municipality (2025).

TECHNOLOGY STACK

• Programming Fundamentals: Proficient in Python and C++; skilled in Linux; capable of configuring CUDA environments and deploying Git/Docker containers.
• Deep Learning: Familiar with PyTorch and HuggingFace ecosystems; able to deploy vLLM; proficient in BitsAndBytes model quantization and AMP mixed-precision acceleration.
• Large Models: Familiar with Transformer and MoE architectures; able to build LangGraph agents; proficient in multimodal model alignment and Prompt projects.